

Student Name: Stephen Kerry Student ID: 20106900

Project Title: Stephen's Smart Doorbell – IoT Assessment Proposal

For this assessment I would like to make a smart doorbell similar to how the ring doorbell works, but also with the use of raspberry pi. It detects motion then captures an image. It will then provide the user with a dashboard of the image captured

It will consist of 4 IOT Layers, a sensor layer, where the motion detector detects movement near the door. The processing layer where the raspberry pi sends data to a server or cloud platform. Then a gateway layer that uses MQTT to a cloud service then an application layer where it shows the image on a web based dashboard.

Hardware:

Hardware needed. Raspberry Pi, Raspberry pi camera module. PIR Motion sensor

Software:

Linux/Python. Flask for web dashboard

MQTT

<https://github.com/StephenKerry/IOTAssesment.git>